

Joao Gabriel de Souza Pinto

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EXPERIENCE

Senior GenAI Software Engineer

May 2025 – Present

Deloitte – Disruption Office (DO)

Remote, Brazil

- Architected and developed the multi-agent orchestrator for **TaxBrain**, one of Deloitte's flagship Tax assets — a high-investment platform helping enterprise clients navigate Brazil's 2026 tax reform. Built the core chatbot using **Agents and RAG** to retrieve and surface specific regulatory changes and adjustments from the new tax legislation.
- Built the agent orchestration layer using **LangGraph and Flask**, enabling dynamic multi-agent workflows deployed on **Kubernetes (AWS EKS)** for scalable, production-grade inference.
- Redesigning the existing orchestrator architecture used across the Disruption Office by implementing **Model Context Protocol (MCP)**, achieving greater control over tool routing, prompt management, and agent behavior.
- Delivered 2 invited talks on GenAI applications: one for **PRODAM** (Sao Paulo City IT Agency) and one at the **Jardim Digital** event, representing Deloitte's AI innovation practice.

AI Researcher

Feb 2025 – Present

CEIA – Centro de Excelencia em Inteligencia Artificial

Remote, Brazil

- Led end-to-end development of a **multi-agent WhatsApp chatbot**, configuring the full architecture from Meta WhatsApp Business API integration to production deployment.
- Designed and implemented an **event-driven architecture** using AWS Lambda, SQS (FIFO queues), and **Agent Development Kit (ADK)**, following bus-based messaging patterns for reliable message processing.
- Managed the complete infrastructure setup including webhook configuration, message routing, and real-time agent orchestration on AWS serverless services.

NLP Data Scientist

Feb 2024 – Feb 2025

Comsentimento

Remote, Brazil

- Integrated open-source clinical LLMs (**BioMistral, Meditron**) into the production pipeline, enhancing automated analysis of clinical study criteria within medical notes.
- Fine-tuned embedding models to optimize the semantic search system, **increasing information retrieval accuracy** for clinical trial matching.
- Designed and deployed evaluation methods for model outputs with a real-time **monitoring dashboard**, improving pipeline reliability and accelerating decision-making.
- Contributed to **Com+Care**, a WhatsApp chatbot leveraging multi-agent architecture (LangGraph) to assist patients with clinical study queries and surface relevant ClinicalTrials.gov research.

NLP Researcher

Oct 2023 – Feb 2025

HAILab-PUCPR – Health Artificial Intelligence Lab

Remote, Brazil

- Led the fine-tuning of LLMs (**LLaMA 7B, Mixtral 7B**) using LoRA, QLoRA, and full training on **Vertex AI**, producing the first LLM trained on Brazilian clinical data.
- Developed evaluation techniques with clear metrics for accuracy, relevance, and clinical text generation quality, establishing benchmarks for Portuguese clinical NLP.
- Published “**MED-LLM-BR: Developing Resource-Efficient Clinical LLMs for Brazilian Portuguese**” at BRACIS 2024 (Springer Nature, DOI: In press, 2025), winning **3rd place Best Paper Award**.

Junior Data Scientist

Sep 2022 – Feb 2024

Cadastra

Remote, Brazil

- Built a **RAG-based Q&A assistant** for internal onboarding, enabling efficient retrieval and synthesis of information from documents, PDFs, and knowledge bases.
- Developed AI-powered content production tools using **LangChain agents** to generate product descriptions, blog articles, and image-based content aligned with client brand identities for clients including Samsung, Vivara, and Gerdau.

EDUCATION

Universidade Federal de Goiás (UFG) – AKCIT/Embrapii <i>Postgraduate in Software Engineering: Automation and Innovation with Generative AI</i>	Goiania, Brazil 2025 – Present
Pontificia Universidade Catolica do Parana (PUC-PR) <i>BTech in Big Data Technology and Analytics</i>	Londrina, Brazil Sep 2020 – Apr 2023

PUBLICATIONS

MED-LLM-BR: Developing Resource-Efficient Clinical LLMs for Brazilian Portuguese	Jan 2025
J. G. S. Pinto et al. Published by Springer Nature in the Intelligent Systems book series (BRACIS 2024). Introduced the first Brazilian Clinical LLM leveraging diverse clinical datasets and advanced fine-tuning techniques (LoRA, QLoRA) with robust validation. DOI: In press, Copyright Year 2025.	

AWARDS AND RECOGNITION

Best Paper Award – 3rd Place BRACIS 2024 (Brazilian Conference on Intelligent Systems)	2024
Lead author of award-winning paper selected among hundreds of submissions at Brazil’s premier AI conference.	
Intel Software Innovator Program – Highlight of the Month	2024
Recognized by Intel for developing a quantized GGUF version of the clinical LLM in collaboration with Alessandro de Oliveira, enabling efficient edge deployment of healthcare AI models.	

TECHNICAL SKILLS

Languages: Python, SQL, Node.js
LLM/AI Frameworks: LangChain, LangGraph, Hugging Face Transformers, PyTorch, TensorFlow, PEFT, vLLM, Llama-Factory, ADK, FastMCP
Techniques: RAG, Fine-tuning (LoRA, QLoRA, SFT), Pre-training, Prompt Engineering, Embeddings, Semantic Search, LLM Routing, Multi-Agent Orchestration, MCP (Model Context Protocol)
Cloud/Infrastructure: AWS (Lambda, SQS, EKS, EventBridge, Bedrock), GCP (Vertex AI, Cloud Run), Kubernetes, Docker
Databases: PostgreSQL, Vector Databases (pgvector, ChromaDB, Qdrant)
Tools: Azure DevOps, GitHub, Slack, Notion, Terraform, Flask, FastAPI
Spoken Languages: Portuguese (Native), English (Highly Proficient)